

Sequence Mining TV Viewing Data Using Embedded Markov Modelling

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Abstract—The advancement of Internet Protocol television (IPTV) has brought significant advantages to TV users in terms of the TV content viewing and control over the Electronic Program Guide (EPG) system, which has provided richer TV viewing behaviour data to gain insights from. The Set-Top-Box (STB) in an IPTV system typically collects user behaviour data as well as monitors system performance in the form of event logs. Sequence mining technologies have been widely used in the study of event logs, such as mining the most frequent items. This paper presents the analysis of event logs on an IPTV box (BT YouView Box), focusing on customers' navigation paths on the EPG system. We propose a new methodology by utilising embedded discrete-time Markov Chains (DTMC) and dynamic programming, to generate maximum likelihood trajectories for customer's navigation sequences. This paper has demonstrated that the DTMC fits well into event logs and has revealed insights into customers' navigation paths. The analysis has also identified several potential areas for further work.

Keywords—Sequence Mining, TV Customer Behaviour, Discrete-Time Markov Chains (DTMC), Process Mining, Clickstream, Dynamic Programming, Python.

I. INTRODUCTION

With the advancement of the Internet and related technologies, there has been a boom in the number of Internet Protocol Television (IPTV) users, as IPTV can provide home users with high-definition live TV broadcast, video-on-demand (VoD) and other interactive services. Therefore, IPTV has been gradually replacing traditional digital cable TVs, satellite TVs, and becoming the most popular TV technology on the market. Meanwhile, along with the rapid development of IPTV services, the importance of accurately understanding users' usage habits of IPTV and digging into the potential needs of IPTV users has become increasingly prominent. On one hand, conducting behaviour analysis on IPTV users can help service providers adjust IPTV service scheduling arrangements (content, timetables), to improve users' viewing experience. On the other hand, they can also provide service providers with decision support, such as advertising pushes, video pushes, etc., thereby increasing the operator's revenue.

There were several studies about IPTV user behaviour early on [1][2], indicating that user behaviour characteristics

in the IPTV scenario not only involve group audience ratings, but also can be described from multiple dimensions such as online duration, channel switching features, time-frequency distribution and other personalisation characteristics. As an important quantitative indicator in the era of the "attention economy", the application of user behaviour study plays a significant role regarding IPTV channel value evaluation, effective advertising, accurate content recommendation, cloud media resource allocation, and fault detection for service providers. To analyse IPTV user behaviour, in general, necessitates the collection of data (including access frequency, access method, access content, etc.) generated during the use of services by users, employing techniques such as statistics, data mining, data analytics, machine learning, to obtain the patterns of how users use the service. Subsequently, prior findings will provide data support for the follow-up development, optimisation, and marketing activities of the service. Moreover, in the era when the Internet has been popularised by thousands of households and user behaviour data is growing explosively, the research work on analysing massive user behaviour data, apart from the IPTV area, also lies in various fields (e.g. network security risk prevention[3], Internet search engine service optimization[4], e-commerce marketing[5], and asocial network research [6]).

However, despite user behaviour being explored by a lot of previous work, there is limited number of recent studies focused on the behaviour of TV audiences [7], especially from the perspective of how TV users interact with the Electronic Program Guide (EPG) system, thereby enabling us to characterise each TV viewer by his viewing habit (i.e. patterns of sequential paths). Additionally, common patterns of TV user's behaviour on the EGP system contain underlying suggestions for IPTV service providers to improve the user interface (UI) and service personalisation. Such data regarding interactions between users and the IPTV system is, in most cases, stored in the form of sequential event-log files. To get insights into data like that, data mining, specifically sequence mining (also known as sequential pattern mining) is the most suitable technique. It has been widely used in many areas for finding frequently occurring ordered events or sub-sequences as patterns. In the IPTV area, sequence mining can provide a way of TV user behaviour analysis by calculating the most

common paths on the EPG system, which could be a great support for companies to evaluate their IPTV services and UI experience. In practice, there have been a number of sequence mining algorithms concerning various application scenarios and PrefixSpan [8] is one of the most popular algorithms on the list. In this paper, we have investigated several novel approaches to the analysis of user behaviour in multiple domains, as well as applications of sequence mining techniques. Based on that, we have developed a method for sequence mining which will generate the maximum likelihood trajectories in IPTV users' navigation log when they are using the EPG on the STB. The proposed method incorporated discrete-time Markov Chains (DTMCs) and a dynamic programming approach since the time-independent transitions in a DTMC model can fit well with the dynamic programming concept. Additionally, our method can also generate the top k ($k > 1$) maximum likelihood trajectories from IPTV users' navigation sequences on the EPG system. The remainder of the paper is organised as follows. Section II is the background of our study and outlines related literature. In section III, we introduce our proposed methodology developed in the study, formulating the problem of generating optimal paths with DTMC. Section IV presents the results using our proposed method. Section V concludes with a summary.

II. BACKGROUND

BT YouView STB provides IPTV services such as live or on-demand TV content across multiple IPTV platforms, including BT Player, Netflix, iPlayer and NOW. It comes with a remote control which offers users sufficient control over the EPG system by clicking buttons for 'tuning into a linear channel', 'watching Video-on-Demand (VoD) programme', 'purchasing/renting a movie', 'subscribing to a package', etc. Such actions are recorded and stored as event logs by the YouView layer on the STB. In addition, other event logs (e.g., power on, power off, network connection) in terms of system performance on the TV box are also stored. High volumes of customer journey data are captured from all STBs on a daily basis, and it is believed that rich data relating to customer journeys while navigating the TV experience can provide new insights to measurably improve engagement. In that regard, BT TV Analytics team has collaborated with Ulster University via the BTIC project, leveraging experience in statistics, sequence modelling and pattern mining etc., to define intelligent methodology/schema and provide comprehensive analysis results to understand customer navigation paths on the EPG which finally lead to a BT channel. Therefore, we have conducted a literature review of sequence mining techniques as well as applications in various domains.

Agrawal et al. [9] first introduced the concept of sequential pattern mining (SPM), referring to the data mining process of finding frequent sub-sequences or items as patterns among a large amount of sequential data. Sequential data refers to data where the order matters, e.g., DNA sequences. Time-series data are usually regarded as sequential data, as they are ordered by timestamps [10] (e.g., user retweets, ATM money withdrawals, patients visiting hospitals, people watching TV), and can be formulated as event sequences in the continuous time domain. In general, continuous event sequences can be formulated as sequences with fixed intervals [11]. Patterns are

basically the key information in sequential data, they refer to a set of items, subsequences or substructures that frequently occur together (or are strongly correlated). It usually contains intrinsic and important knowledge in real-world datasets, e.g. a specific segment of DNA sequences. Based on that, in recent studies of sequential event data, there has emerged another concept - episode rule. An episode rule refers to event patterns which can predict the occurrence of consequent events based on the occurrence of antecedent events [12]. An episode is generally a set of ordered sequential events. Fig. 1 gives an example for episode rule mining in sequential data, where capital letters denote events and numbers denote timestamps.

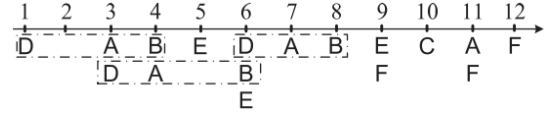


Fig. 1. Example of an episode rule in sequences [12]

As is shown in Fig. 1, episode $\langle D, A, B \rangle$ occurs three times, $\langle D, A \rangle \xrightarrow{\Delta t} \langle B \rangle$ is an episode rule if we take $\langle B \rangle$ as a consequent event. The time interval Δt between $\langle D, A \rangle$ and $\langle B \rangle$ is usually given in the episode rule mining process. Ao et al. [12] introduced new concepts of fixed-gap episode as well as precise-positioning episode rule (PER), and they proposed tree structure-based algorithms to solve the precise-positioning episode rules (MIPER) problem. An example of a PER in Fig. 1 is $\langle D, A \rangle \xrightarrow{2} \langle E, F \rangle, \langle 3 \rangle$, where numeric value 3 is the newly proposed constant in episode mining models, referring to the interval ("gap") between consequent events E and F. They argued that the Tries [14] structure significantly improved the performance of their method in terms of computation efficiency and memory saving. Laxman [13] proposed efficient and memory-saving algorithms for counting frequencies of non-overlapped episodes to find most frequent episodes in data streams. Many types of research on frequent episode mining (FEM) have been carried out, yet there has been no attempt to consider the time interval between consequent events in episode rule mining process while many real-world applications requires much time sensitivity.

Typically, sequential pattern mining from sequences is a critical research direction in data mining. It is an effective knowledge discovery process as patterns contain inherent information. Yu et al. [10] proposed two algorithms respectively based on the existing Apriori and PrefixSpan algorithms, to solve the problem of mining sequential patterns from d -dimensional sequences, where d is greater than 2. PrefixSpan [8] is one of the most popular algorithms on the list for sequential patterns mining. It can detect patterns efficiently in a prefix-projected pattern growth method. Saraf et al. [15] introduced a PrefixSpan-based approach for finding sequential patterns on large datasets efficiently. They introduced a new parameter - maximum prefix length (MPL) in the PrefixSpan algorithm to set the length of the prefix pattern, facilitating running the algorithm on large datasets. PrefixSpan was also applied by Hijawi and Saheb [16] for making comparisons with a newly proposed method - data stream sequence pattern mining (DSSPM). They proved that DSSPM could provide more efficient calculation using accurate top frequent patterns. To analyse web logs for obtaining user behavioural patterns, Weichbroth [17] proposed an Apriori-like frequent web

pattern (FWP) algorithm. The algorithm was based on the Apriori algorithm which is for frequent item sets mining over transactional databases. Additionally, [18] has provided an overview of current web data mining categories (namely, content, structure and usage), techniques and relevant tools.

In the IPTV area, research on user behaviour can be categorised into two major perspectives. One category is focused on the programme and content information TV user are watching, while the other mainly concerns TV users viewing behaviour, namely, the habit of interacting with IPTV EPG systems. The former topic of research has achieved success in recommender systems [19][20], while the latter one has led to studies on TV customer churn prediction [21], channel switch prediction [22], etc. However, studies addressing both types of information are often preferred and in high demand in the IPTV industry. For instance, to analyse IPTV users' viewing behaviours regarding timestamp and topic of the programme, Chen et al. [23] proposed a coupled latent Dirichlet allocation model (cLDA) which combined both topics and temporal information from a video on demand (VoD) system. The model could discover popular interests among users, reflected by the probability distribution of activity patterns, known as patterns of programme topics. Another recent study [24] in IPTV area has applied frequent item set mining approaches to generate TV user's fingerprints for identification. Most importantly, they revealed that "human activities in cyberspace naturally form temporal sequences, and the transitions between consecutive activities can also be used as fingerprints for user identification." Built upon this finding, our work aims to characterise users by their 'maximum likelihood trajectory' in their sequential event logs.

The size of data coupled with the variation in patterns presents further challenge for sequence mining for user behaviour analysis. Markov modelling is an attractive solution. This is mainly because the states of the Markov Chain are independent of each other, which is consistent with the characteristics of user behaviour sequential logs. Markov Modelling has been widely applied to many domains [25]. In an early pioneering study of user behaviours in one of the largest cellular networks in the US, Gerber et al. [26] applied statistical tools like Markov Model and tri-nonnegative matrix factorisation (tNMF), to investigate data usage patterns among network users regarding data consumption, applications and devices used. Liu et al. [27] built up a queueing system to model customer behaviour of both abandonment and recurrent consumption with a Markov model to formulate the staffing problem in a call system. In other applications such as web browsing logs analysis [28], HTTP request link prediction [29] and news recommendation [30], Markov Modelling has played a vital part in the study, incorporating the power of probability and mathematics to gain insights into sequences.

In summary, despite previous studies on IPTV user behaviour analysis and a wide application of sequential pattern mining techniques, extensive research on IPTV user behaviour in terms of navigation paths on the EPG system coupled with the programme information is still in great demand. Also, Markov Modelling in IPTV area has great potential since it can facilitate the analysis by calculating probabilities which can save time and memory compared to other algorithms.

III. METHODOLOGY

YouView's Data and Insights team have prepared and shared an anonymised dataset with YouView client logs for the customer behavioural data analysis and modelling purposes. It is primarily about events occurred during a customer's TV journey on the STB, including clickstreams generated by the remote control, systematic events measuring the STB performance and other diverse information. Our work is mainly focused on a subsampled dataset from it for testing purposes. This section will describe our proposed sequence mining method step-by-step with the utility of a DTMC model, from data manipulation to the calculation of maximum likelihood trajectories in event sequences.

A. Data preparation

One of the challenges in IPTV data analysis is the complexity of data streams. In our study, the dataset originally consists of tangled streams of events such as system events, user navigation path, etc, which creates a lot of noise (e.g. irrelevant data streams for our study). 'EVENT ID' attribute in the data which indicates the type of event logs on the STB. With BT's prior knowledge about event IDs, that column can assist us with teasing out target streams of our interest. There are roughly 4 categories of event IDs in the raw data: 1) UI change events – generally refer to the navigation path on the IPTV system, with URLs; 2) Specific records of 4 IPTV services usage – specifying Linear DTT channels, IP channels, VoD, Personal Video Recorder (PVR); 3) Common events across those 4 services (e.g. application start/complete/...); 4) System event logs – STB boot info, power manager activity, software installation, etc. In our case, we are more interested in a user's clickstream on the STB, and their usage pattern of IPTV services. We also need breakpoint identifiers in the event stream to define a user's TV journey from the start to the end, so we include STB boot information in our target dataset. Additionally, there is an extra column in the original table indicating channel information and genre, but mostly for the IP channel service. To simplify our study, our focus will be the analysis of the IP channel service, specifically. So, we select the specific data stream by event ID to get a lighter dataset.

B. Event labelling

In the lighter dataset, there are several columns for each event record, two of them containing key information about navigation paths on the EPG system: 'EVENT ID' and 'UI_CONTEXT'. One indicates the event type, the other shows the exact URL of the current page on the EPG system. The EPG system on STB provides a long list of channels as well as other content which are all embedded in a hierarchical menu, and the URLs are the accurate locator showing where the user navigates through. So, a series of URLs represent the navigation path of the user on the EPG system. However, content under either columns are initially long text. For example, one sample event ID is 'ui.tunerequested', and a sample URL is 'livetv/main-menu/guide/action-panel/dialog', which are very comprehensive but not so reader-friendly for our study. In order to retain information while making it more human readable, we apply mapping tables (namely, 'Event Index Table', 'UI Context Table') which store the original information as key and our manually generated labels as

values to convert long text into short labels. The function is implemented in Python, and the process for one example is shown in Figure 2 below. In this way, we combine event type information with the corresponding URL, to give each event record a simplified look, also providing an easier starting point for sequence generation from events.

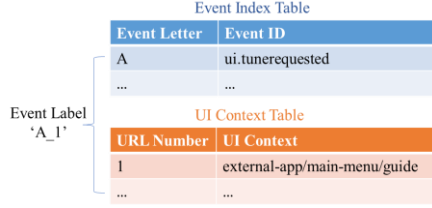


Fig. 2. Example of event labelling

C. Sequence generation

Based on event labels, along with timestamp information, we are able to convert time-series events into sequences, aggregated by each TV journey. Here we have applied specific scenarios to define a sensible TV journey for the analysis: “1. Customers start with turning on the IPTV system, walking through the menu a bit, and finally choose an IP channel service; 2. The IP session is more than three minutes; 3. To filter events prior to each three minutes IP session, only the last block of events following a gap of 10 minutes or more is kept.” The purpose is to narrow down the focus of the use case in our study, paying more attention to user’s navigation paths right before they start watching something. We have implemented the sequence generation function in Python with the application of scenarios. An example of the functionality as well as the input & output format is shown in Fig. 3.

User ID	Event	Timestamp
1	a	0
1	b	1
1	c	2
1	e	3
2	b	0
2	f	1
2	g	2
2	e	3
3	a	0
3	c	1
3	e	2

→

User ID	Event Sequence
1	<a,1,b,1,c,1,e>
2	<b,1,f,1,g,1,e>
3	<a,1,c,1,e>

Fig. 3. Example of sequence generation

Fig.3 shows that sequential events with timestamp are converted into sequences following “event-interval-event” format which contains original information from the dataset. All values in both tables are for demonstration purposes only. A snapshot of the real data after sequence generation function is shown in Figure 4 below.

```
SEQUENCE_CONTENT
K_385,6.479,D_58,0.482,K_386,0.119,K_385,13575.646,K_386
A_71,0.134,K_385,30.293,A_58,389.318,D_58,0.55,K_386
```

Fig. 4. A snapshot of generated sequences

In Fig.4, sequences are stored under the column ‘SEQUENCE_CONTENT’. Intervals are calculated to three decimal places. Each sequence refers to a TV journey before the customer starts an IP session more than 3 minutes. Other information is also kept for the sequence, such as starting and ending point, duration of the journey, channel genre, etc. The generated sequence table, especially the format of ‘event-interval-event’ transitions forms up a good foundation for the next step in our study, namely Markov Modelling.

D. Embedded-DTMC Modelling

A continuous-time Markov Chain (CTMC) [31] is a continuous stochastic process where for each state, the process

will change state according to an exponential random variable and then move to a different state as specified by the probabilities of a stochastic matrix. While inside a CTMC, an embedded DTMC can be obtained by looking at a semi-Markov process which is only about transition instances. In other words, we ignore how long the process stays in a state, and only focus on the probabilities of making transitions from each state to all the other states when a transition occurs. In our case, the transitions between events in each sequence and the duration of the transition provide key information about a continuous stochastic process in real life, which can fit a CTMC model. However, since our focus is on the navigation path in terms of the order of user visiting different menu pages on the EPG system, the information about transition duration isn’t important in this particular study. Only the transition between events is focused and subsequently processed using an embedded-DTMC model.

A DTMC is a discrete time and memoryless discrete event stochastic process i.e. a sequence of random processes $\{X_0, X_1, X_2, \dots, X_t\}$, where X_t is the state at time t . The distribution of the forthcoming state X_{t+1} depends only on the current state X_t and does not depend on previous states $X_0, X_1, X_2, \dots, X_{t-1}$ [32]. The state space of a DTMC is a set S composed of all possible state values that each X_t can take, e.g. $S = \{s_0, s_1, s_2, \dots, s_r\}$. The change between states in the chain is named either “transition” or “step”, and the probability p_{ij} to move from state s_i to state s_j in one step is the transition probability [32]:

$$p_{ij} = \Pr(X_1 = s_j | X_0 = s_i) \quad (1)$$

In the embedded-DTMC, the state space is the full set of all distinct event labels in sequences. Since we have the genre type information for IP sessions, we divide sequences by genre and apply a DTMC for each. It is expected to see some similarities or differences in the results between genres. The transition matrix for each DTMC was defined by using the maximum likelihood estimator (MLE) [32] of the p_{ij} entry, where n_{ij} stands for the number of pairs $(X_t = s_i, X_{t+1} = s_j)$ found in sequences. The calculation for each p_{ij} MLE estimator is shown in Equation (2) [32]:

$$\hat{p}_{ij}^{MLE} = \frac{n_{ij}}{\sum_{u=1}^k n_{iu}} \quad (2)$$

Also, in each step X_t , the distribution over the states can be represented by a row vector $\mathbf{x} = \{x_i\}$, $(\sum_i x_i = 1, x_i \geq 0)$. When $t = 0$, we call $\mathbf{x}$ an initial state vector, denoted by:

$$\mathbf{\pi}_0 = [Pr(X_0 = s_1), Pr(X_0 = s_2), \dots, Pr(X_0 = s_r)] \quad (3)$$

Subsequently we calculate the initial state vector [32] for each DTMC by the Equation (4), which refers to the initial probability distribution over all states when $t = 0$.

$$Pr(X_0 = s_i) = \frac{count(X_0 = s_i)}{count(X_0)} \quad (4)$$

Additionally, for our DTMC models, there is one absorbing state [32] by default, which is the event indicating the start of an IP session in our scenario. In the next step, we will propose an algorithm using dynamic programming to find the maximum likelihood trajectory through a DTMC, with a given initial vector and an absorbing state.

E. Maximum likelihood trajectory through DTMC

The transient states are $S_1, \dots, S_k$ with absorbing (final) state S_{k+1} , initial $1 \times k$ probability vector $\boldsymbol{\alpha} = \{\alpha_i\}$ and $(k+1)$ by $(k+1)$ transition matrix $\mathbf{T} = \{t_{ij}\}$.

Let $f_n = \{f_n(i)\}$ be the vector of n -step maximum likelihood trajectories, where $f_n(i)$ is the n -step maximum likelihood trajectory starting from state S_i for $i=1, \dots, n$. Correspondingly, $s_n = \{s_n(i)\}$ is the vector of first steps in this trajectory. Then, using Bellman's Principle of Optimality, we have initial conditions: $f_1(i) = t_{i, n+1}$ for $i=1, \dots, n$ and $s_1(i) = n+1$ for $i=1, \dots, n$ and

$$f_{n+1}(i) = \max_j \{t_{ij} f_n(j)\} \quad (5)$$

$$s_{n+1}(i) = \arg\max_j \{t_{ij} f_n(j)\} \quad (6)$$

We solve these recursive equations until all the elements of f_n converge. The maximum likelihood trajectory then has probability:

$$p_{\max} = \max_i \{\alpha_i f_{n_{\text{con}}}(i)\} \quad (7)$$

where n_{con} is the final converged value of n for f_n .

The corresponding maximum likelihood trajectory is found by recursively computing the s_n vectors.

To calculate the **second** maximum likelihood trajectory, the probability is denoted by $P_n^{\max(N-1)}$:

$$P_n^{\max(N-1)} = \max_{N-1} \{\alpha_i f_n(i)\}, n \in (0, n_{\text{con}}] \quad (8)$$

We continue finding the largest value of $P_n^{\max(N-1)}$ until convergence by the given initial state vector α , then subsequent states are re-calculated recursively from the s_n vectors. Similarly, to find the top k maximum likelihood trajectory ($k > 1$), it is just required to find the largest value of $P_n^{\max(N-k)}$ and the corresponding path. We have implemented the algorithm in Python code and calculated top 10 maximum likelihood trajectories for the DTMC model of each genre.

IV. RESULTS

Based on the generated sequence table, for each IP channel genre, we have implemented DTMC modelling and stored relevant information about the Markov Model including the transition matrix, states set, initial state vector in csv files. Subsequently by following the steps in section III-E, we have generated the top 10 maximum likelihood trajectories in the navigation sequences for each genre. There are 8 major genres of IP channels on the BT TV box. However, to avoid intellectual property violations against BT or YouView, we denote them by numbers only. For the top-10 maximum likelihood trajectories, we link events back to their original text through the mapping table and visualise each event with a distinct colour. The visualisation of the results for each genre is displayed in Figure 5.

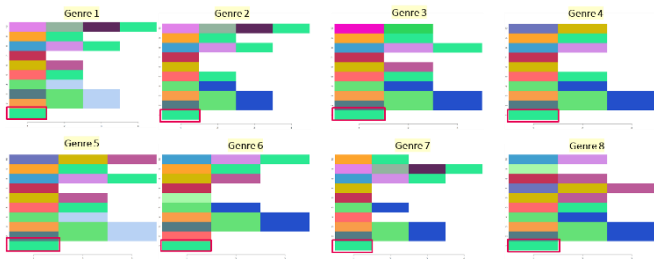


Fig. 5. Top 10 maximum likelihood trajectories for each Genre

The visualisation for each genre, presented in Fig. 5, from top to bottom, shows the top 10 most-likely trajectories with probabilities in ascending order, which means the bottom

sequence pattern is most likely to occur (with the highest probabilities). Each coloured unit cell refers to a TV journey event regarding either a UI or Service (dtt,ip,vod,pvr) event, and the colour is unique for each event, specifying the event information. The results show that across all genres, the maximum likelihood path for users to enter an IP session greater than 3 mins is to use a direct tune event (green coloured cells at the bottom). After looking into the details of the coloured patterns in sequences across genres, we come up with more preliminary results based on the graph:

- Most genres have  except genre 8, which in fact refers to URLs on the EPG system as 'external-app -> livetv'. We can infer that people watching genre 7 follow a relatively different path such as a preferred shortcut on the remote control.
- Similarly, most genres have  except genre 8, where that colour refers to the URL 'livetv/main-menu/mytv/recordings/action-panel'. It can indicate that people are not likely to navigate through recording panel to watch channels of genre 8.
- Most genres have  except 6 and 7. The sequence refers to URLs 'livetv/main-menu -> livetv', which indicates people watching genre 6 or 7 are not likely to go through the live TV menu before they start the IP session.
-  and  are shared across genres except genre 1 and genre 5. Those two similar sequences both end with 'livetv->dtt_end', which indicates that before people tuning into genre 1 or genre 5 channels, they are not likely to finish a linear DTT channel session.
- Only genre 6 and genre 8 have the sequence  which is 'livetv/search', this can imply that people tend to use the search function to watch IP channels of genre 6 and 8.
- Genre 1 and genre 5 have similar patterns in the sequences; genre 2,3,4 and 8 resemble each other in sequence patterns, indicating that people follow similar navigation paths to watch those genres.

V. CONCLUSION AND FUTURE WORK

In this paper, we have proposed and applied a methodology for the analysis of IPTV customer's navigation paths on the EPG system, from the perspective of sequence mining. DTMCs and dynamic programming approaches were incorporated to model log files on the STB provided by YouView team. We followed several steps in carrying out our study, including data cleaning, event labelling, sequence generation and DTMC modelling, etc. DTMC modelling turned out to fit very well in our scenario, as customer's navigation behaviour is a good example of stochastic process.

We have also gained preliminary results from top-10 maximum likelihood trajectories in customers' navigation paths when watching IP channels of 8 different genres. Such results extracted using our methodology were interpreted to reveal patterns of high probabilities in customer behaviour on the EPG system, some of these have provided interesting insights into navigation paths related to genre types. We would

like to carry out more in-depth comparison between sequence patterns of different genres for our future work. One potential direction is to implement DTMC transition matrix chi-square testing to see more details about the similarities of TV journey behaviours between genres, to complement the current approach and results.

The dataset used for our work so far is relatively small, including only 100 customers. The proposed methodology and obtained results will serve as the basis in transferring the knowledge to bigger dataset from 10,000 BT customers. Most importantly, in the future work, we aim to compare the results generated by our approach with the results from the traditional sequence mining algorithm PrefixSpan, so that we can validate our approach. Besides, the temporal factor that we have left aside will be involved in our future work, to apply CTMCs. We also intend to characterise customers by their navigation paths (e.g. 'fast navigator', 'wanderer'), potentially to apply HMM and detect different hidden states along customer's TV journeys. Our methodology is also suitable for other analyses of business process data for sequential events such as ATM withdraw events, bank transactions, hospital visiting records, etc, in terms of process mining or sequence mining.

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